Machine Learning (2026) 115:238
https://doi.org/10.1007/s10994-026-07177-w

On Kernel Eigen-Alignments of KRR: Reconstruction and Generalization

Yang Liu[1] · **Ernest Fokoue**[2] · **Richard Lange**[1] · **Daniel Krutz**[1]

Received: 19 June 2025 / Revised: 19 September 2026 / Accepted: 23 September 2026
© The Author(s) 2026

Abstract
This paper investigates the critical role of Eigenalignments between the kernel matrix and learning targets in achieving robust generalization in learning problems. We establish a direct connection between generalization performance in kernel methods and the estimation of Eigenvectors and Eigenvalues of matrices, offering a more intuitive understanding compared to prior work with minimal assumptions. We also show that, since the prediction task in KRR is essentially the weighted sum of Eigenvectors/singular vectors, by analyzing how much error can be caused by perturbations to the kernel matrix, we can then derive a bound on this generalization error using the estimation stability of matrix Eigenvalues and Eigenvectors. Compared with previous work, our analysis concentrates on finite-sample settings and on the generalization error arising from having a suboptimal finite training set. Our findings reveal that in kernel methods, as long as the kernel is of high rank, the near-zero reconstruction error can be trivially obtained, implying that the reconstruction error will have limited predictive power for generalization. Additionally, we establish a generalization bound from an Eigenvalues/Eigenvectors estimation perspective, showing that strong generalization requires increasing Eigenvector alignment, Eigenvalue magnitude, or gaps between consecutive Eigenvalues. Finally, we also demonstrate empirically our simplified risk estimator is able to track testing errors in trailing Eigendirections more closely than the existing literature.

Keywords Learning theory · Eigen-alignment · KRR

1 Introduction

This paper investigates the critical role of Eigenalignments between the kernel matrix and learning targets in achieving satisfactory generalization performance in learning problems. These Eigenalignments, which represent the inherent properties of the data and the selected kernel, are frequently overlooked in conventional learning theory. We establish a direct connection between generalization in kernel ridge regression and the estimation of Eigenvec-

Editors: Michelangelo Ceci and Qibin Zhao.

Extended author information available on the last page of the article

tors and Eigenvalues. Our work is based on the foundational work of Canatar et al. (2020); Jacot et al. (2006); Simon et al. (2021). Compared to previous work, our analysis focuses on the finite sample regime, thus avoiding issues related to the slow convergence rate that has been pointed out by Simon et al. (2021); Wei et al. (2022). Based on a novel perspective of matrix perturbation theory, our analysis makes minimal assumptions and provides a more intuitive and clearer understanding of the mechanism of Eigenalignments through the lens of Eigenvectors/Eigenvalues estimations, revealing previously hidden or unnoticed implications. We show that for a given kernel matrix, the generalization error can depend non-trivially on the learning targets. Additionally, in high dimensions, non-parametric kernels can achieve nearly zero reconstruction error, rendering the relationship between reconstruction error and generalization error negligible in these cases. We demonstrate this using synthesized learning targets, as classical machine learning datasets tend to contain well-behaved labels, which can lead to spurious correlations. For a given kernel matrix with enough rank, we demonstrate that achieving a near-zero training loss is nearly always feasible. This finding implies that the training error has limited utility in predicting the generalization error.

Furthermore, we establish a generalization bound based on the predictive estimation of finite sample Eigenvalues and Eigenvectors. We show that a good generalization requires either an increase in the alignment of Eigenvectors, an increase in the Eigenvalue, or a widening of the gap between adjacent Eigenvalues. Our results show that the regularization parameter often has a minimal impact. This suggests that its primary role is to mitigate numerical instability, rather than its traditionally perceived function of reducing hypothesis space in the non-parametric space. When regularization does influence generalization, it is due to its alignment with trailing Eigenvectors, occurring when the corresponding Eigenvalue is comparable in value to the regularization parameter. We show that the regularization is more related to salvaging a bad performance than improving a good performance. Hence, we can summarize this using the common bias-variance terminology: bias is attributed to the inability of the top Eigenvectors to represent the learning targets, while variance arises from the requirement for the learning targets to be represented by the trailing Eigenvectors. These findings align with the existing literature *e.g.* Hastie et al. (2022); Liang and Rakhlin (2018). Our findings also suggest that overfitting in the non-parametric kernel ridge regression setting is essentially related to the alignments of trailing Eigenvectors, which are dominated by noise. As a result, this implies the true nature of overfitting, in this case, is the kernel lacking the capacity to differentiate noise and information due to either the data being fundamentally too noisy or the kernel choice not being optimal.

Our analysis emphasizes the effects of the choice of learning targets on the model's generalization behavior, particularly in the context of Kernel Ridge Regression (KRR). Figure 1 presents two key aspects of our findings: we demonstrate that the generalization error (on a logarithmic scale) increases as the rank of the Eigenvector with which the learning targets are aligned increases. Regularization, on the other hand, only has a slightly positive impact (reducing the generalization error) when the learning targets are aligned with trailing Eigenvectors. The details of the experiments can be found in Sect. "4.1".

1.1 Problem Setup

In this paper, we study the kernel selection problem in the context of large-dimension kernel ridge regression, which is also referred to as kernel LS-SVM in the literature. Our work is

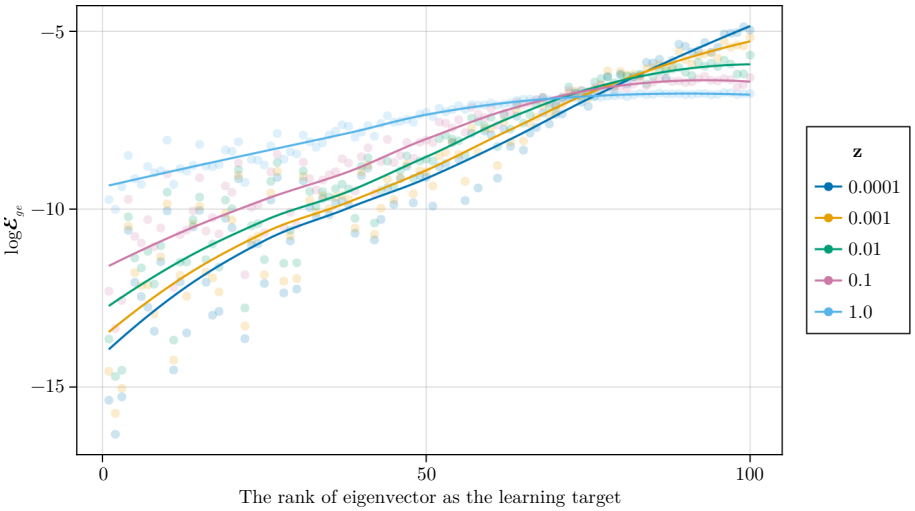


Fig. 1 Learning targets of the task affect the generalization performance. Learning targets aligned with top Eigenvectors (lower rank) tend to have lower generalization errors. Regularization tends to have a negative impact on generalization error for most learning targets, except those aligned with trailing ones (high-rank Eigenvectors)

in the same vein as recent research in kernel alignment *e.g.* Jacot et al. (2006); Canatar et al. (2020).

We consider the typical setting of Kernel Ridge Regression (KRR). Denote the training dataset $\mathcal{D} = \{X, y\}$ of size n and let K_n represent the kernel constructed from the input data sample X. Each entry in K_n is evaluated using $\mathrm{k}(x_i, x_j)$, where a kernel function $\mathrm{k} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$, K_n represents the $n \times n$ matrix of kernel evaluations $\mathrm{k}(x_i, x_j)$, and KRR learns a function that maps any given x to a prediction based on a weighted sum of kernel evaluations on the training data:

$$\hat{f}_n(x) = \sum_{i=1}^{n} \alpha_i \, \mathrm{k}(x, x_i) \,,$$

where $(\alpha_1, \ldots, \alpha_n)^\top = (K_n + zI_n)^{-1} y$ and $z \in \mathbb{R}^+$ is the ridge regularization parameters. Although our analysis can be applied to all kernel functions, our focus is leaning using inner product kernels in high-dimension cases, as we consider them to be more relevant in modern machine learning settings. As shown in Couillet and Benaych-Georges (2016); Couillet and Liao (2021); Liao and Couillet (2019), inner product kernels are more effective than distance-based kernels such as Gaussian kernels: distance-based kernels minus the diagonal has an asymptotic rank of one. By inner product kernels, we specifically refer to a kernel matrix calculated using

$$K_{i,j} = h(x_i^\top x_j),$$

where h is some chosen function. We will refer to the training data labels as y learning targets and will use $\hat{y}$ to denote the predictions or reconstructions of the training data, which can be expressed simply as

$$\hat{y} = K_n(K_n + zI_n)^{-1}y\,.$$

The reconstruction error, also known as training loss, is

$$\mathcal{E}_{re} := \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2,$$

where y_i denotes the ith label for y. For any pair of observed data $\{\tilde{X}, \tilde{y}\}$, the predictions are

$$\hat{\tilde{y}} = \tilde{K}(K_n + zI_n)^{-1}y\,,$$

where $\tilde{K}_{i,j} = \mathrm{k}(\tilde{x}_i, x_j)$. We are particularly interested in how the learning target affects the generalization error, which is also quantified using the expected squared error:

$$\mathcal{E}_{ge} := \mathbb{E}(\tilde{y}_j - \hat{\tilde{y}}_j)^2,$$

where $(\tilde{x}_j, \tilde{y}_j)$ is an unseen data point and $\hat{\tilde{y}}_j = f(\tilde{x}_j)$. Furthermore, in our analysis, we will extensively utilize the spectral decomposition of the kernel matrix K_n, in the sense that $K_n = \sum_{i=1}^{n} d_i u_i u_i^\top$. We will assume that the Eigenvalues are sorted $d_1 \geq d_2 \geq \cdots \geq d_n$. We refer to the first few (small-number) Eigenvalues and Eigenvectors, characterized by small values, as the top Eigenvalues and top Eigenvectors, whereas the remaining Eigenvalues and Eigenvectors are termed trailing Eigenvalues and trailing Eigenvectors.

We list the most used notation in Table 1.

Table 1 List of notations used

Notation	Meaning
K_n	$n \times n$ Kernel matrix
$\mathrm{k}(\cdot,\cdot)$	Scalar kernel function
u	Eigenvector of the kernel matrix
d	Eigenvalue of the kernel matrix
z	Regularization parameter
x, y	Training input data point and learning targets
$[\cdot]_i$	ith data point
$[\cdot]^*$	Optimal decorator
$[\hat{\cdot}]$	Prediction decorator for (x, y)
$\tilde{x}$, $\tilde{y}$	Testing data pair
$\mathcal{E}_{re}$	Reconstruction error
$\mathcal{E}_{ge}$	Generalization error
$\tilde{\mathrm{k}}_{\tilde{x}_j}$	Row vector from the kernel matrix corresponding to $\tilde{x}_j$
$\mathbb{P}$	Projection operator

2 Main Results

2.1 Summary Of Results and Contributions

We first provide a brief summary of the main results presented in our paper, delegating detailed discussions to the subsequent sections. In the KRR setting, the reconstructions can be written in terms of Eigenvalues and Eigenvectors of K_n:

$$\hat{y} = \sum_{i=1}^{n} u_i (d_i u_i^\top y)/(d_i + z)\,.$$

In other words, training data reconstructions are simply weighted sums of Eigenvectors (Canatar et al., 2020; Jacot et al., 2006; Simon et al., 2021). In equation (3), we will then introduce a novel decomposition of the reconstruction error:

$$\|y - \hat{y}\|_n^2 = n\mathcal{E}_{re} = \sum_{i:d_i \neq 0} \frac{z^2 (u_i^\top y)^2}{(d_i + z)^2} + \sum_{i:d_i = 0} (u_i^\top y)^2.$$

where $\|\cdot\|_n$ is the empirical norm of L_2. We observe that the minimization of the reconstruction error $\mathcal{E}_{re}$ is not correlated with the accuracy of estimating the Eigenvalues and Eigenvectors. This implies that reconstruction is a rather trivial task in high-dimensional spaces, and the generalization performance of KRR cannot be accurately predicted by the training error alone, in contrast to the methods of Jacot et al. (2006).

We proceed to demonstrate that the key to the generalization error (as opposed to the reconstruction error) lies in an accurate estimation of Eigenvalues and Eigenvectors. Specifically, we show that for any new data point $\tilde{x}_j$, when the learning targets are aligned with the top Eigenvectors of the kernel matrix, the predictive estimation error can be bounded by

$$C_j \sum_{i=1}^{n} |\mathbb{P}^y_{u_i^*}| \left(\sum_{k:k\neq i} \frac{2\|\Delta K_n\|}{|d_i^* - d_k^*|} + (1 - \Theta_i) + \frac{\|\Delta K_n\| + 2 + \sqrt{2 - 2\Theta_i}}{d_i^* + z} \right),$$

where K_n^* is the "optimal kernel matrix" of size n. Here, $\|\cdot\|$ for the matrix is understood as the operator norm, $\{d_i^*, u_i^*\}$ represents the Eigenpairs of K_n^*, C_j is a multiplier that depends on the new data point $\tilde{x}_j$ but not on the training data, and Θ_i denotes the estimation alignment of the ith Eigenvector (which will be introduced formally later). For any given training set, K_n^* and K_n are fixed. Hence, we notice that in order to achieve good generalization, we need either to increase the alignment of Eigenvectors, increase the Eigenvalue d_i^*, or increase the Eigen gap $|d_i^* - d_k^*|$; all of these can be achieved if the learning targets are aligned with the top Eigenvectors, *i.e.* $\mathbb{P}^y_{u_i^*} := u_i^{*\top} y = 0$ for all i except for several top Eigenvectors u_i^*.

2.1.1 Comparison to Previous Work

There have been many works on the topic of Eigen-alignments of KRR; compared to past work, our paper's unique contributions are the following:

1. Previous work such as Simon et al. (2021); Jacot et al. (2006) makes Gaussian assumptions to obtain the results: this assumption likely not true in practice (Barzilai and Shamir, 2023). Our results require no such distributional assumptions.
2. Methodologically, we develop a perturbation-based analysis on the kernel alignments that allows us to derive non-asymptotic finite sample bounds. This analysis is facilitated by a novel method of constructing an optimal finite-sample dataset.
3. We obtain explicit bounds on generalization error in terms of Eigen-alignments and spectral gaps (Theorem 1) and suggest a simple empirical estimator eq.(12). Previous literature contains either convoluted formalization such as Jacot et al. (2006), or hard-to-interpret implicit formulations of generalization error such as Canatar et al. (2020); Bordelon et al. (2020), obfuscating the underlying mechanisms of generalization. As a result, several key insights are buried in the literature, and some formulations were even misinterpreted by the authors themselves:

   (a) Contrary to claims of Jacot et al. (2006), we show that reconstruction error or training error has minimal predictive power for generalization performance.
   (b) We emphasize that ridge regularization has minimal effects in improving generalization beside the controlling numeral issues.
   (c) We highlight these effects in cases of kernel-task misalignment, i.e. cases where targets align with trailing Eigenvectors.

2.2 Eigen-Characterization of KRR

We first present a straightforward yet insightful Eigen-characterization of the KRR. While the results resemble earlier findings in the literature, such as Jacot et al. (2006); Canatar et al. (2020); Simon et al. (2021), we provide a more intuitive interpretation and offer new insights. The goal is to understand the training and generalization behavior of KRR in a non-parametric regime, particularly in high dimensions. For this purpose, recall the reconstruction operator for $z \in \mathbb{R}^+$,

$$A = K_n(K_n + zI_n)^{-1}.$$

then $\hat{y} = Ay$. Through algebraic manipulation, we have

$$\begin{aligned} A &= K_n(K_n + zI_n)^{-1} \\ &= K_n(K_n + zI_n)^{-1} \\ &= U\Lambda U^\top \sum_{i=1}^{n} \frac{u_i u_i^\top}{d_i + z} \\ &= \sum_{i=1}^{n} \frac{d_i u_i u_i^\top}{d_i + z} \end{aligned}$$

where $\Lambda = \mathrm{diag}(d_1, \ldots, d_N)$; thus, for training predictions

$$\begin{aligned}\hat{y} =& Ay = \left(\sum_{i=1}^{n} \frac{d_i u_i u_i^\top}{d_i + z}\right) y \\ =& \sum_{i=1}^{n} \frac{d_i u_i u_i^\top y}{d_i + z} = \sum_{i=1}^{n} \frac{(d_i \mathbb{P}_{u_i}^y) u_i}{d_i + z}\end{aligned} \tag{1}$$

where P is the projection operator such that $\mathbb{P}_{u_i}^y := u_i^\top y$. Therefore, predictions are the weighted sum of Eigenvectors, with weight $(d_i \mathbb{P}_{u_i}^y)/(d_i + z)$. The alignment of the target vector suggests the magnitude of the contribution; the regularization coefficient z on the other hand, determines the directions of the contribution. Poor alignment of the Eigenvector with the target vector, despite the large Eigenvalue of the kernel matrix, can result in a negligible contribution to the final predictions. Additionally, this implies that the target vector can learn at most the linear combinations of the Eigenvectors of the kernel matrix. This result resembles Proposition 11 of Muandet et al. (2016).

Since $\{u_i\}_i^n$ forms an orthonormal basis, $y = \sum_i \mathbb{P}_{u_i}^y u_i$ and $d_i/(d_i + z)$ actually adjust the weights to ensure that the predictions are more dependent on the dominant Eigenvectors. This approach avoids reliance on purely noisy Eigenvectors and incorporates reconstruction loss along with generalization error. *However, this non-rigorous intuition is not precise. We will rigorously analyze both reconstruction and generalization later.*

Remark 1 (Regularization) In the non-parametric KRR setting,

$z > 0$ helps to regularize the model by favoring Eigenvectors that correspond to large Eigenvalues; this is in the same spirit as PCA: when $d_i + z \gg d_i \mathbb{P}_{u_i}^y$, the corresponding Eigenvector will be negligible. This also resembles the idea of signal capture threshold from Jacot et al. (2006): if the signal along an Eigenvector direction is too weak compared to the corresponding Eigenvalue, the signal will not be learned by the model.

Remark 2 When $z = 0$, all Eigenvectors contribute equally; however, when $z > 0$, the Eigenvectors corresponding to larger Eigenvalues receive greater relative weights. Fine-tuning z therefore initially seems to help informative Eigenvectors stand out. However, as shown above, z can be negligible in $d_i + z \gg d_i \mathbb{P}_{u_i}^y$ if d_i is large in this non-parametric regime. Conversely, z has a non-negligible effect only when the labels are aligned with the trailing Eigenvectors, indicating a less than ideal learning scenario. This finding is consistent with existing literature, such as Liang and Rakhlin (2018); Hastie et al. (2022), suggesting that regularization is not essential to achieve good performance.

2.3 Reconstruction Error

Following our Eigen-characterization of the KRR, we further analyze the reconstruction error.

Proposition 1 *According to the non-parametric KRR assumptions, the reconstruction error of y, with respect to the kernel matrix K_n, is*

$$\mathcal{E}_{re} := \frac{1}{n}\|y - \hat{y}\|_n^2 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{\mathbb{P}_{u_i}^y z}{d_i + z}\right)^2 \tag{2}$$

where $z \in \mathbb{R}^+$ is the regulation parameter, $\mathbb{P}_{u_i}^y := u_i^\top y$ and $\{u_i\}_{i=1}^n$ are the Eigenvectors of the kernel matrix K_n.

Proof Following our previously established notations, we have

$$\begin{aligned}\hat{y} - y &= \sum_{i=1}^{n}\frac{(d_i\mathbb{P}_{u_i}^y)u_i}{d_i + z} - y = \sum_{i=1}^{n}\frac{d_i u_i \mathbb{P}_{u_i}^y}{d_i + z} - \sum_{i=1}^{n} u_i u_i^\top y \\ &= \sum_{i=1}^{n}\frac{d_i u_i \mathbb{P}_{u_i}^y}{d_i + z} - \sum_{i=1}^{n} u_i \mathbb{P}_{u_i}^y = -\sum_{i=1}^{n}\frac{z u_i \mathbb{P}_{u_i}^y}{d_i + z}\end{aligned}$$

Then, reconstruction error is

$$\begin{aligned}\mathcal{E}_{re} &= \frac{1}{n}(\hat{y} - y)^\top(\hat{y} - y) \\ &= \frac{1}{n}\left(\sum_{i=1}^{n}\frac{z u_i \mathbb{P}_{u_i}^y}{d_i + z}\right)^\top\left(\sum_{j=1}^{n}\frac{z u_j \mathbb{P}_{u_j}^y}{d_j + z}\right) \\ &= \frac{1}{n}\sum_{i=1}^{n}\left(\frac{\mathbb{P}_{u_i}^y z}{d_i + z}\right)^2.\end{aligned}$$

□

Remark 3 Although it may be tempting to consider that the reconstruction error is minimal when $y \perp U$ (the Eigenvector matrix), this is not valid. By construction, the columns of U span the space of $\mathbb{R}^n \ni y$. In fact, for a given y, the quantity in the numerator, $\sum_{i=1}^{n}(\mathbb{P}_{u_i}^y)^2 = \|y\|^2$, is independent of the choice of the kernel. Hence, minimizing the reconstruction error is a matter of choosing a kernel that d_i is large for whichever modes u_i are most aligned with y Bordelon et al. (2020); Simon et al. (2021).

According to Proposition 1, for the KRR to have a small reconstruction error, the adjusted projection magnitude should be small compared to the corresponding adjusted Eigenvalue (by the regularization parameter z). We observe that the reconstruction error can be minimized in the following sense.

Proposition 2 *(Reconstruction Error) In non-parametric KRR, the reconstruction error can be decomposed as*

$$\mathcal{E}_{re} = \sum_{i:d_i\neq 0}\frac{z^2(\mathbb{P}_{u_i}^y)^2}{(d_i + z)^2} + \sum_{i:d_i=0}(\mathbb{P}_{u_i}^y)^2. \tag{3}$$

If $\mathbb{P}_{u_i}^y = 0$ for $i \in \{i : d_i = 0\}$, then for any $\varepsilon > 0$, the reconstruction error $\mathcal{E}_{re} < \varepsilon$ cor-

responds to some regularization parameter z.

Proof From eq.(2), we have

$$\frac{1}{n}\sum_{i=1}^{n}\left(\frac{\mathbb{P}_{u_i}^{y} z}{d_i+z}\right)^2 = \frac{1}{n}\left(\sum_{i:d_i\neq 0}\left(\frac{\mathbb{P}_{u_i}^{y} z}{d_i+z}\right)^2 + \sum_{i:d_i=0}\left(\frac{\mathbb{P}_{u_i}^{y} z}{d_i+z}\right)^2\right)$$
$$= \frac{1}{n}\left(\sum_{i:d_i\neq 0}\left(\frac{\mathbb{P}_{u_i}^{y} z}{d_i+z}\right)^2\right)$$

The last equality holds due to our assumption $\mathbb{P}_{u_i}^{y} = 0$ for $i \in \{i : d_i = 0\}$. Since

$$\lim_{z\to 0}\frac{1}{n}\left(\sum_{i:d_i\neq 0}\left(\frac{\mathbb{P}_{u_i}^{y} z}{d_i+z}\right)^2\right) = 0,$$

the claim holds. □

Remark 4 In theory, as long as the Eigenvalue $d_i > 0$, we can find some z, such that $\forall \varepsilon > 0, \mathcal{E}_{re} < \varepsilon$. However, in practice, this is limited by the machine precision or numerical error.

Proposition 2 shows that given the learning targets, as long as the rank of the kernel matrix permits, the reconstruction error can be arbitrarily small by reducing the regularization parameters. This suggests a disconnect between the reconnection error and the generalization error, as illustrated below.

Corollary 1 *(Triviality of Reconstruction) An arbitrary full-rank kernel matrix K_n can reconstruct any learning target y with $\mathcal{E}_{re} \approx 0$. Furthermore, the larger the Eigenvalues, the smaller the reconstruction error $\mathcal{E}_{re}$.*

Proof Due to Proposition 2, we know that the full-rank kernel matrix can reconstruct any learning target y, since $d_i \neq 0, \forall i \in \{i : \mathbb{P}_{u_i}^{y} \neq 0\}$. It is also clear that larger d_i will lead to smaller $\mathcal{E}_{re}$ based on eq.(2).

□

Before we attribute the above phenomenon to overfitting, we can make the following observations: (1) the inaccuracy in estimating the Eigenvectors and Eigenvalues from the sample does not necessarily result in poor reconstruction error; (2) as long as $d_i \gg z$, the corresponding contribution to the reconstruction error will remain near zero; (3) the Eigencomponents corresponding to zero Eigenvalues cannot be reconstructed. This demonstrates that reconstruction in non-parametric KRR is relatively a trivial task; in some loose sense this resembles the 1-nearest neighbor predictor. As long as the kernel matrix is of high rank, we can achieve near-zero reconstruction (training) error. The error is primarily a limitation imposed by the numerical stability associated with the use of small z. The triviality of

reconstruction arises from the perfect alignment of the training and predicting kernels in the reconstruction process. It is worth mentioning that this is purely an algebraic fact, independent of data distributions.

We will also demonstrate empirically in Sect. "4.1.2", in the non-parametric KRR that the near-zero reconstruction error is not necessarily indicative of overfitting. The actual quality of generalization depends on more nuanced factors, which will be analyzed in the following section.

2.4 Generalization Error and Predictive Eigenalignments

The previous section showed that the training or reconstruction error in non-parametric KRR is connected to Eigenvalues and Eigenvectors of the sample kernel K_n, as well as the alignment of these Eigenvectors with the training targets. Next, we will demonstrate how the generalization error depends on how *accurately* those sample Eigenvalues and Eigenvectors are estimated from a given finite data set.

For a finite sample size, not all Eigenvectors and Eigenvalues can be estimated equally well. It is known that with a finite sample size, the top Eigenvalues and Eigenvectors can be approximated more accurately in terms of relative error[1] (see also *e.g.* Mestre (2008); Loukas (2017)). This suggests that there are two distinct sources of error: (1) a kind of "bias" due to the inability to represent the target solely by the informative Eigenvectors (those with large Eigenvalues, which are more easily estimated from finite data); and (2) a kind of "variance" due to the necessity of representing the target with the non-informative Eigenvectors (those with small Eigenvalues, which are not easily (or feasible) estimated from finite data which will cause "confusion" in predictions).

2.4.1 Eigen-Characterization of Generalization

Generalization is the question of how well any specific data sample $\mathcal{D}$ of size n can approximate other data samples $\tilde{\mathcal{D}}$ drawn from the same distribution. We derive a novel expression for the generalization error in terms of the relative errors of the Eigenvalue and Eigenvector estimates. Intuitively, we notice that the generalization behavior of a non-parametric Kernel Ridge Regression (KRR) model depends on the extent to which the kernel matrix is affected by the samples used in the training process. If the predictions are highly dependent on volatile information, such as trailing Eigenvectors, then the generalization error is likely to be large. To this end, we utilize existing tools in matrix analysis to examine how the different samples affect the Eigenvectors. *We first illustrate this idea non-rigorously.*

For each individual test point $\tilde{x}_j$, we can write the prediction as

$$\hat{\tilde{y}}_j = \sum_{i=1}^{n} \frac{\mathbb{P}^{y}_{u_i} \mathbb{P}^{u_i}_{\hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}}}{d_i + z},$$

where $\hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}$ is the jth row of $\tilde{K}$ corresponding to the new data point $\tilde{x}_j$. To better understand predictive behavior, it is worth exploring how predictions for generalization differ from those for reconstruction.

[1] The actual dynamics of estimation accuracy are more nuanced, but it is outside the scope of this paper.

To this end, we can assume that the vector $\hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}$ comes from a row of the kernel matrix $\tilde{K}$ with the same training size, *i.e.* $n \times n$ (this allows spectral decomposition as it pertains to the training kernel matrix). We also temporarily overlook the fact that the test example $\hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}$ exists solely in the test kernel $\tilde{K}$, and we allow decomposition $\tilde{K} = \sum_{i=1}^{n} \tilde{d}_i \tilde{u}_i \tilde{u}_i^\top$. We have

$$\begin{aligned}
\hat{\tilde{y}}_j &= \sum_{i=1}^{n} \frac{\mathbb{P}_{u_i}^{y} \mathbb{P}_{\hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}}^{u_i}}{d_i + z} \\
&= \sum_{i=1}^{n} \frac{\mathbb{P}_{u_i}^{y} [\sum_k^n \tilde{d}_k \tilde{u}_k \tilde{u}_k^\top u_i]_j}{d_i + z} \\
&= \sum_{i=1}^{n} \frac{\mathbb{P}_{u_i}^{y} [\tilde{d}_i \tilde{u}_i \tilde{u}_i^\top u_i + \sum_{k \neq i} \tilde{d}_k \tilde{u}_k \tilde{u}_k^\top u_i]_j}{d_i + z}
\end{aligned}$$

where $[\cdot]_j$ denotes the selection of the jth element of the vector. We can then make several observations: (1) the sample size of the training set determines the size of the hypothetical kernel matrix for the testing data. This is consistent with the fact that the size of the testing data should not affect the test accuracy; (2) Small $\tilde{u}_k^\top u_i$ for $i \neq k$ and large $\tilde{u}_k^\top u_i$ for $i = k$ will lead to a correct prediction, as both are approaching their respective optimal counterparts.

2.4.2 Generalization Bounds

To analyze the generalization error, we restrict ourselves to considering the target function $f \in \mathbb{H}$, where $\mathbb{H}$ represents the Reproducing Kernel Hilbert Space (RKHS) corresponding to the Mercer kernel k. Following the standard convention (see *e.g.* Cucker and Zhou (2007)), we can write $\mathrm{k}(x_j, x_k) = \sum_i \sqrt{d_i} \psi_i(x_j) \psi_i(x_k)$, where $\{d_i, \psi_i\}$ represent the Eigenvalues and Eigenfunctions of a linear operator induced by k. Since scaled Eigenfunctions $\{\sqrt{d_i}\psi_i\}$ form an orthonormal basis in $\mathbb{H}$, we can express $f = \sum_i a_i \psi_i$. For any finite training sample of size n, we have a predictive function $\hat{f}_n \in \mathbb{H}$ obtained via kernel ridge regression. By the representer theorem, $\hat{f}_n = \sum_i^n \alpha_i \mathrm{k}(x_i, \cdot)$, where $\alpha = (K_n + zI)^{-1} y \in \mathbb{R}^n$. As we increase the sample size n, we observe

$$\lim_{n \to \infty} \hat{f}_n = f,$$

where $f \in \mathbb{H}$ corresponds to the ground truth generalization error minimizer. Then, for the new test point $\tilde{x}_j$, let $\hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}$ be the row vector (of length n) of the kernel matrix $\tilde{K}$ that corresponds to $\tilde{x}_j$. Additionally, $\tilde{\mathrm{k}}_{\tilde{x}_j} \in \mathbb{H}$ is the kernel map that corresponds to $\tilde{x}_j$. We can express the prediction for $\tilde{x}_j$ as

$$\tilde{y}_j = f(\tilde{x}_j) := \langle \tilde{\mathrm{k}}_{\tilde{x}_j}, f \rangle_{\mathbb{H}}, \text{ and } \hat{\tilde{y}}_j = \hat{f}_n(\tilde{x}_j) := \langle \hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}, (K_n + zI_n)^{-1} y \rangle .$$

Moreover, the predictive error is

$$\hat{y}_j - \tilde{y}_j = \langle \hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}, (K_n + zI_n)^{-1} y \rangle - \langle \tilde{\mathrm{k}}_{\tilde{x}_j}, f \rangle_{\mathbb{H}}.$$

The accuracy of the prediction $\hat{y}_j = \hat{f}_n(\tilde{x}_j)$ depends on the empirical estimator $\hat{f}_n \in \mathbb{H}$ —whose coefficients $\boldsymbol{\alpha} = (K_n + zI_n)^{-1} y \in \mathbb{R}^n$ are derived via the Representer Theorem—providing a tight approximation of the true target function $f \in \mathbb{H}$ with respect to the RKHS norm. Consequently, this convergence requires that the empirical kernel operators constructed from the finite sample evaluations (represented by K_n and $\mathrm{k}_{\tilde{x}_j}$) sufficiently concentrate around the true integral operator and covariance structures of the underlying data-generating distribution as $n \to \infty$.

Our key idea is to compare the learned predictor $\hat{f}_n$ not directly to the ground-truth function f, but rather to a hypothetical best predictor, f_n^* which is learned from only n samples, where the samples themselves are selected to minimize the generalization error of f_n^*. That is, we define f_n^* as the KRR solution to the hypothetical optimal data sample $\{x_1^*, \ldots, x_n^*\}$, with the values of x^* selected to minimize generalization error. Note that this will require the order of the optimal data points to be "aligned" with the order of the actual data sample in a sense that we will make precise below.

Let K_n^* be the kernel matrix for these n optimal data points, and let $\{d_i^*, u_i^*\}$ be the Eigenvalues and Eigenvectors of K_n^*. The optimal $\tilde{\mathrm{k}}^*_{\tilde{x}_j}$ and the predictive $\hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}$ (utilizing an actual non-optimal training set) also have their representations with corresponding Eigenvectors:

$$\tilde{\mathrm{k}}^*_{\tilde{x}_j} = \sum_{i=1}^{n} \beta_i^* u_i^*, \text{ and } \hat{\tilde{\mathrm{k}}}_{\tilde{x}_j} = \sum_{i=1}^{n} \hat{\beta}_i u_i,$$

recall that $\{d_i, u_i\}$ are the Eigenpairs of the actual training kernel matrix K_n. Since $\hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}$ depends on the arbitrary input $\tilde{x}_j$, which determines $\hat{\beta}_i$, an accurate value of $\hat{\beta}_i$ will result in an accurate prediction for $\tilde{x}_j$. To this end, we have

$$\begin{aligned}
&|\hat{\tilde{y}}_j - \tilde{y}_j| \\
&= |\langle \hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}, (K_n + zI_n)^{-1} y \rangle - \langle \tilde{\mathrm{k}}_{\tilde{x}_j}, f \rangle_{\mathbb{H}}| \\
&= \left| \left(\langle \hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}, (K_n + zI_n)^{-1} y \rangle - \langle \tilde{\mathrm{k}}^*_{\tilde{x}_j}, f_n^* \rangle \right) + \left(\langle \tilde{\mathrm{k}}^*_{\tilde{x}_j}, f_n^* \rangle - \langle \tilde{\mathrm{k}}_{\tilde{x}_j}, f \rangle_{\mathbb{H}} \right) \right| \\
&= \left| \left(\left\langle \sum_{i=1}^{n} \hat{\beta}_i u_i, \sum_{i=1}^{n} \frac{\mathbb{P}^y_{u_i}}{d_i + z} u_i \right\rangle - \left\langle \sum_{i=1}^{n} \beta_i^* u_i^*, \sum_{i=1}^{n} \frac{\mathbb{P}^y_{u_i^*}}{d_i^* + z} u_i^* \right\rangle \right) + \left(\langle \tilde{\mathrm{k}}^*_{\tilde{x}_j}, f_n^* \rangle - \langle \tilde{\mathrm{k}}_{\tilde{x}_j}, f \rangle_{\mathbb{H}} \right) \right| \\
&\leq \left| \sum_{i=1}^{n} \hat{\beta}_i \frac{\mathbb{P}^y_{u_i}}{d_i + z} - \sum_{i=1}^{n} \frac{\mathbb{P}^y_{u_i^*}}{d_i^* + z} \beta_i^* \right| + \left| \langle \tilde{\mathrm{k}}^*_{\tilde{x}_j}, f_n^* \rangle - \langle \tilde{\mathrm{k}}_{\tilde{x}_j}, f \rangle_{\mathbb{H}} \right| \\
&\leq \underbrace{\left| \sum_{i=1}^{n} \hat{\beta}_i \left(\frac{\mathbb{P}^y_{u_i}}{d_i + z} - \frac{\mathbb{P}^y_{u_i^*}}{d_i^* + z} \right) \right|}_{\text{Term1}} + \underbrace{\left| \sum_{i=1}^{n} \frac{\mathbb{P}^y_{u_i^*}}{d_i^* + z} \Delta\beta_i \right|}_{\text{Term2}} + \underbrace{\left| \langle \tilde{\mathrm{k}}^*_{\tilde{x}_j}, f_n^* \rangle - \langle \tilde{\mathrm{k}}_{\tilde{x}_j}, f \rangle_{\mathbb{H}} \right|}_{\text{Term3}},
\end{aligned} \tag{4}$$

where we assume the orthonormality of the Eigenvectors and define $\Delta\beta_i = \hat{\beta}_i - \beta_i^*$. Observing eq. (4), "Term 1" captures the error caused by incorrect Eigenvalue estimation; "Term

2" captures the error caused by the bias of the training set; and "Term 3" captures the error caused by the finite sample size.

Bounding the generalization error amounts to bounding the first two terms in eq. (4), as the last term is fixed. To this end, we will consider the finite sample kernel matrix of the training dataset as a perturbed version of the optimal sample kernel matrix since our performance depends on the finite estimation of the Eigenfunctions. We can then utilize the tools from matrix perturbation theory to develop our bound on eq. (4).

We first introduce a well-known lemma that will aid us in developing our results.

Lemma 1 *(Davis-Kahan Davis and Kahan (1970); Eldridge et al. (2017)) Suppose X and ΔX are symmetric matrices, and let the perturbed matrix be defined as $\tilde{X} = X + \Delta X$. Let (λ_i, u_i) and $(\tilde{\lambda}_j, \tilde{u}_j)$ denote the Eigenvalue-Eigenvector pairs of X and $\tilde{X}$, respectively. If we define the spectral gap for a given index t as $\delta_t = \min_{j \neq t} |\tilde{\lambda}_j - \lambda_t|$, then the principal angle θ_t between the unperturbed Eigenvector u_t and the perturbed Eigenvector $\tilde{u}_t$, defined by $\theta_t = \cos^{-1}(|u_t^\top \tilde{u}_t|)$, satisfies the inequality:*

$$\sin\theta_t \leq \frac{\|\Delta X\|}{\delta_t}$$

where $\|\Delta X\|$ denotes the operator norm of the perturbation matrix ΔX.

It is clear that δ_t is large for isolated Eigenvalues and small for trailing Eigenvalues. While there are tighter bounds than Lemma 1, such as Eldridge et al. (2017); Fan et al. (2018), these bounds are generally more involved in their formulation; for the brevity of the result, we focus on Lemma 1 in our paper.

We next demonstrate that, given a fixed sample kernel error $\Delta K_n = K_n - K_n^*$, aligning learning targets with the top Eigenvectors will result in improved generalization performance. Our analysis aims to demonstrate how the generalization performance is affected by the perturbations. To this end, for a fixed kernel difference ΔK_n, we have

$$|u_i^\top u_i^*| \geq \sqrt{\max\left\{0, 1 - \left(\frac{\|\Delta K_n\|}{\delta_i}\right)^2\right\}} =: \Theta_i \tag{5}$$

since u_i and u_i^* are unit vectors, $\delta_i = \min\{|d_k - d_i^*| : i \neq k\}$.

2.4.3 The Upper Bound of Term 1 in eq. (4)

To bound term 1, suppose $d_i = d_i^* + \Delta d_i$, we have

$$\begin{aligned}
&\frac{\mathbb{P}^y_{u_i}}{d_i + z} - \frac{\mathbb{P}^y_{u_i^*}}{d_i^* + z} \\
&= \frac{\mathbb{P}^y_{u_i}(d_i^* + z) - \mathbb{P}^y_{u_i^*}(d_i + z)}{(d_i + z)(d_i^* + z)} \\
&= \frac{(d_i^* + z)(\mathbb{P}^y_{u_i} - \mathbb{P}^y_{u_i^*}) - \Delta d_i \mathbb{P}^y_{u_i^*}}{(d_i + z)(d_i^* + z)} \\
&= \frac{(d_i^* + z)\langle u_i - u_i^*, y\rangle - \Delta d_i \mathbb{P}^y_{u_i^*}}{(d_i + z)(d_i^* + z)}.
\end{aligned} \tag{6}$$

To gain a sense of the concentration behavior of the above expression. We make use of the following lemma, which is modified from Loukas (2017).

Lemma 2 *(Theorem 3.2 of Loukas (2017)) For Hermitian matrices K and $\tilde{K} = \Delta K + K$, with Eigenvectors u_i and $\tilde{u}_j$ respectively, we have*

$$|\langle u_i, \tilde{u}_j\rangle| \leq \frac{2\|\Delta K\|}{|d_i - d_j|}$$

provided $d_i \neq d_j$, and the perturbation is sufficiently small such that the condition

$$\text{sign}(d_j - d_i)2\tilde{d}_i > \text{sign}(d_j - d_i)(d_i + d_j)$$

holds true.

Remark 5 Compared to the original Theorem 3.2 in Loukas (2017), we have adapted the Lemma 2 with our notations and adjusted the right-hand side based on the simple fact that $\|\Delta K_n u_i\|_2 \leq \|\Delta K_n\|$.

Lemmas 1 and 2 both contain bounds in terms of ΔK_n, and Lemma 2 in particular requires that g564 K and $\tilde{K}$ (which will be K_n and K_n^* in what follows) are "close." There are two ways in which $\Delta K_n = K_n - K_n^*$ may be large. First, the optimal $\{x_1^*, \ldots, x_n^*\}$ may not be unique. Second, the *order* of training data must be aligned with the order of the optimal data, i.e. x_i should be close to x_i^* for all i. We do not specify K_n^* directly; instead, we assume that, among all possible permutations of data and among all possible sets of optimal data points, we take K_n^* to be whichever minimizes misalignment to K_n. We do not think this is a strong assumption; the condition $\text{sign}(d_j - d_i)2\tilde{d}_i > \text{sign}(d_j - d_i)(d_i + d_j)$ in Lemma 2 is known to hold asymptotically almost surely for large Eigengaps and sufficiently large sample sizes (Loukas, 2017). *We will assume this in our further analysis, as it further supports our claim that when the learning targets align with the trailing Eigenvectors, errors cannot be bounded.*

To bound eq.(6), we first observe that through straightforward algebra, we have

$$
\begin{aligned}
u_i - u_i^* &= \sum_k^n \langle u_i, u_k^* \rangle u_k^* - u_i^* \\
&= \sum_{k:i \neq k} \langle u_i, u_k^* \rangle u_k^* + \langle u_i, u_i^* \rangle u_i^* - u_i^*,
\end{aligned} \tag{7}
$$

additionally, we have

$$
\begin{aligned}
&\left| \hat{\beta}_i \left(\frac{\mathbb{P}^y_{u_i}}{d_i + z} - \frac{\mathbb{P}^y_{u_i^*}}{d_i^* + z} \right) \right| \\
&\leq |\hat{\beta}_i| \left| \frac{(d_i^* + z)(\mathbb{P}^y_{u_i} - \mathbb{P}^y_{u_i^*}) - \Delta d_i \mathbb{P}^y_{u_i^*}}{(d_i + z)(d_i^* + z)} \right| \\
&= |b_i|(d_i + z) \left| \frac{(d_i^* + z)(\mathbb{P}^y_{u_i} - \mathbb{P}^y_{u_i^*}) - \Delta d_i \mathbb{P}^y_{u_i^*}}{(d_i + z)(d_i^* + z)} \right| \\
&= |b_i| \left| \frac{(d_i^* + z)(\mathbb{P}^y_{u_i} - \mathbb{P}^y_{u_i^*}) - \Delta d_i \mathbb{P}^y_{u_i^*}}{d_i^* + z} \right| \\
&= |b_i| \left| ((u_i^\top - u_i^{*\top}) y) - \frac{\Delta d_i \mathbb{P}^y_{u_i^*}}{d_i^* + z} \right| \\
&\leq |b_i| \left(|(u_i^\top - u_i^{*\top}) y| + \left| \frac{\Delta d_i \mathbb{P}^y_{u_i^*}}{d_i^* + z} \right| \right),
\end{aligned}
$$

where $b_i := \hat{\beta}_i/(d_i + z)$. Plugging eq.(7) into the last line, we obtain

$$
\begin{aligned}
&\left| \hat{\beta}_i \left(\frac{\mathbb{P}^y_{u_i}}{d_i + z} - \frac{\mathbb{P}^y_{u_i^*}}{d_i^* + z} \right) \right| \\
&\leq |b_i| \left(\left| \left(\sum_{k:i \neq k} \langle u_i, u_k^* \rangle u_k^* + \langle u_i, u_i^* \rangle u_i^* - u_i^* \right)^\top y \right| + \left| \frac{\Delta d_i \mathbb{P}^y_{u_i^*}}{d_i^* + z} \right| \right).
\end{aligned}
$$

Using Lemma 2, we have

$$
\begin{aligned}
&\left| \hat{\beta}_i \left(\frac{\mathbb{P}^y_{u_i}}{d_i + z} - \frac{\mathbb{P}^y_{u_i^*}}{d_i^* + z} \right) \right| \\
&\leq |b_i| \left(\left| \sum_{k:k \neq i} \frac{2\|\Delta K_n\|}{|d_i^* - d_k^*|} \mathbb{P}^y_{u_k^*} \right| + |\langle u_i, u_i^* \rangle u_i^{*\top} y - u_i^{*\top} y| + \left| \frac{\Delta d_i \mathbb{P}^y_{u_i^*}}{d_i^* + z} \right| \right) \\
&\leq |b_i| \left(\left| \sum_{k:k \neq i} \frac{2\|\Delta K_n\|}{|d_i^* - d_k^*|} \mathbb{P}^y_{u_k^*} \right| + \left| \mathbb{P}^y_{u_i^*} (1 - \Theta_i) \right| + \left| \frac{\|\Delta K_n\| \mathbb{P}^y_{u_i^*}}{d_i^* + z} \right| \right),
\end{aligned} \tag{8}
$$

where the last inequality is due to $|\Delta d_i| \leq \|\Delta K_n\|$, which follows from the definition of the operator norm, and eq.(5). Notice that the summation term contributes only when $\mathbb{P}^y_{u_K^*} \neq 0$.

2.4.4 The Upper Bound of Term 2 in eq. (4)

Next, we bound the term 2 on the right-hand side of eq. (4). Since both $\tilde{\mathrm{k}}^*_{\tilde{x}_j}$ and $\hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}$ are the n-dimensional projections of the infinite-dimensional counterpart $\tilde{\mathrm{k}}_{\tilde{x}_j}$, we also have

$$\begin{aligned}\left|\sum_{i=1}^{n}\frac{\mathbb{P}^y_{u_i^*}}{d_i^*+z}\Delta\beta_i\right| &= \left|\sum_{i=1}^{n}\frac{\mathbb{P}^y_{u_i^*}}{d_i^*+z}(\hat{\beta}_i-\beta_i^*)\right| \\ &= \left|\sum_{i=1}^{n}\frac{\mathbb{P}^y_{u_i^*}}{d_i^*+z}\left(\mathbb{P}^{\hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}}_{u_i}-\mathbb{P}^{\tilde{\mathrm{k}}^*_{\tilde{x}_j}}_{u_i^*}\right)\right| \\ &\le \sum_{i=1}^{n}\frac{|\mathbb{P}^y_{u_i^*}|}{d_i^*+z}\|\mathrm{k}\|\left(2+\sqrt{2-2u_i^\top u_i^*}\right).\end{aligned} \tag{9}$$

where the last inequality is due to Cauchy-Schwarz, and $\|\mathrm{k}\| := \max(\|\hat{\tilde{\mathrm{k}}}_{\tilde{x}_j}\|, \|\tilde{\mathrm{k}}^*_{\tilde{x}_j}\|)$. We summarize the analysis presented above in the following theorem.

Theorem 1 *Using previously established notations and within the framework of non-parametric Kernel Ridge Regression (KRR), consider a training set of size n with the kernel matrix represented as K_n. define*

$$\mathfrak{S}_j := C_j\sum_{i=1}^{n}|\mathbb{P}^y_{u_i^*}|\left(\sum_{k:k\neq i}\frac{2\|\Delta K_n\|}{|d_i^*-d_k^*|}+(1-\Theta_i)+\frac{\|\Delta K_n\|+2+\sqrt{2-2\Theta_i}}{d_i^*+z}\right) \tag{10}$$

where C_j is some constant dependent on $\tilde{x}_j$, and $\Delta K_n = K_n - K_n^*$ and K_n^* is the optimal kernel matrix of size n, $\{d_i, u_i\}$ and $\{d_i^*, u_i^*\}$ are the Eigenpairs of K_n and K_n^*, respectively, and Θ_i is defined in eq.(5). Assuming that $2\,\mathrm{sign}(d_i^*-d_k^*)d_i > \mathrm{sign}(d_i^*-d_k^*)(d_i^*+d_k^*)$ holds, the expected squared generalization error

can be bounded as

$$\mathcal{E}_{ge} := \mathbb{E}(\tilde{y}_j-\hat{\tilde{y}}_j)^2 \le \mathbb{E}\left(\mathfrak{S}_j+\left|\langle\tilde{\mathrm{k}}^*_{\tilde{x}_j}, f_n^*\rangle-\langle\tilde{\mathrm{k}}_{\tilde{x}_j}, f\rangle_{\mathbb{H}}\right|\right)^2. \tag{11}$$

Proof Let the estimation error

$$\mathcal{E}_{es}(\tilde{x}_j) := |\tilde{y}_j-\hat{\tilde{y}}_j|+\left|\langle\tilde{\mathrm{k}}^*_{\tilde{x}_j}, f_n^*\rangle-\langle\tilde{\mathrm{k}}_{\tilde{x}_j}, f\rangle_{\mathbb{H}}\right|.$$

We then combine eq. (4), eq.(8), and eq. (9) to obtain (note that $\hat{\beta}_i$ and β_i^* are dependent on $\tilde{x}_j$).

$$
\begin{aligned}
&\mathcal{E}_{es}(\tilde{x}_j) \\
&\leq \left|\sum_{i=1}^{n} \hat{\beta}_i \left(\frac{\mathbb{P}^y_{u_i}}{d_i + z} - \frac{\mathbb{P}^y_{u_i^*}}{d_i^* + z}\right)\right| + \left|\sum_i \frac{\mathbb{P}^y_{u_i^*}}{d_i^* + z} \Delta\beta_i\right| \\
&\leq \sum_{i=1}^{n} |b_i| \left((u_i^\top - u_i^{*\top})y + \frac{|\mathbb{P}^y_{u_i^*}| \|\Delta K_n\|}{d_i^* + z}\right) + \sum_{i=1}^{n} \frac{|\mathbb{P}^y_{u_i^*}|}{d_i^* + z} \|\mathrm{k}\| \left(2 + \sqrt{2 - 2\Theta_i}\right) \\
&\leq \sum_{i=1}^{n} |b_i| \left(\left|\sum_{k:k\neq i} \frac{2\|\Delta K_n\|}{|d_i^* - d_k^*|} \mathbb{P}^y_{u_k^*}\right| + \left|\mathbb{P}^y_{u_i^*}(1 - \Theta_i)\right| + \left|\frac{\|\Delta K_n\| \mathbb{P}^y_{u_i^*}}{d_i^* + z}\right|\right) \\
&\quad + \sum_{i=1}^{n} \frac{|\mathbb{P}^y_{u_i^*}|}{d_i^* + z} \|\mathrm{k}\| \left(2 + \sqrt{2 - 2\Theta_i}\right) \\
&\leq C_j \sum_{i=1}^{n} |\mathbb{P}^y_{u_i^*}| \left(\sum_{k:k\neq i} \frac{2\|\Delta K_n\|}{|d_i^* - d_k^*|} + (1 - \Theta_i) + \frac{\|\Delta K_n\| + 2 + \sqrt{2 - 2\Theta_i}}{d_i^* + z}\right),
\end{aligned}
$$

where in the last inequality, we denote $C_j := \max\{|b_1|, \cdots, |b_n|, \|\mathrm{k}\|\}$. Then, the claim follows naturally. □

Remark 6 Although the eq.(10) might seem complex, its structure is actually quite simple: each term is essentially the projection of the learning targets multiplied by the corresponding estimation error for that Eigenvector. We will verify this empirically in Sect. "4.1.3": the magnitude and volatility of the generalization error increase from the top Eigenvectors to the trailing ones.

Theorem 1 shows a given fixed $\|\Delta K_n\|$, which measures the difference between various kernel matrices formed by each sample of fixed size n. This difference is usually determined by the nature of the data obtained from the sampling process. Our results suggest the following: *(1) given the difference between empirical kernel matrices, the learning targets aligned with the top Eigenvectors can generalize better; (2) alignments with Eigenvectors corresponding to isolated Eigenvalues can further help with generalization.* The latter observation also coincides with the empirical findings from Martin and Mahoney (2020), which the authors noticed that well-generalized neural networks have weights that have isolated Eigenvalues.

Remark 7 Theorem 1 assumes the same regularization between K_n and K_n^*. We notice that by increasing the regularization parameter z, the differences among various kernels will diminish. Additionally, a large z will cause the generalization error to dominate the approximation error. Such a large regularization will cause K_n to behave like an identity matrix, which can lead to near-zero reconstruction but catastrophic generalization, as shown in Sect. "4.2".

Remark 8 Since the top Eigenvectors and Eigenvalues play a crucial role in predictions, and the sample top Eigenvalues and Eigenvectors are known to be inconsistent, it is possible to use shrinkage functions to reduce the estimation error of the kernel matrix. The shrinkage estimation for the kernel matrix has been proposed by Lancewicki (2019).

2.5 A Simple Empirical Estimator

The theoretical error bound for $\mathbb{E}\mathcal{E}_{ge}$ establishes a rigorous guarantee for the estimation error; however, it depends entirely on inaccessible optimal quantities, namely the true Eigenvalues d_i^*, the population Eigenfunctions u_i^*, *etc.* To demonstrate this bound empirically, we construct a simply empirical estimator $\widehat{\mathcal{E}}$ that captures the core structural factors that will influence the predictability of the learning targets.

We replace the unobserved optimal quantity K^* with the empirical one K constructed from the training samples. The unobservable population projection $|\mathbb{P}^y_{u_i^*}|$ is substituted with its empirical counterpart, and dropping other non-computable components, the foundational shape of the estimator becomes a weighted sum of the inverse empirical Eigenvalue gaps:

$$\widehat{\mathcal{E}} \propto \sum_{i=1}^{n} |\mathbb{P}^y_{u_i}| \sum_{k:k\neq i} \frac{1}{|d_i - d_k| + \tau}$$

where $\tau = 1 \times 10^{-3}$ is a small number to prevent dividing by zero. While $\widehat{\mathcal{E}}$ accurately captures the trend and structural stability of the learning problem, its raw magnitude might not match actual estimation error due to dropping non-computable terms. We remedy the magnitude mismatch by a factor based on the learning target:

$$\gamma_n = \frac{\|y\|_2}{n}$$

Hence, our simple empirical error estimator is the following:

$$\widehat{\mathcal{E}} = \frac{\|y\|_2}{n} \sum_{i=1}^{n} |\mathbb{P}^y_{u_i}| \left(\sum_{k:k\neq i} \frac{1}{|d_i - d_k| + \tau} \right). \tag{12}$$

2.6 Relation Between Reconstruction and Generalization

Reconstruction error can be arbitrarily small as long as the learning target lies in the span of the kernel's Eigenvectors (Proposition 2). As we have suggested earlier, this result in non-parametric KRR is algebraic and does not rely on data distribution assumptions. From this viewpoint, overfitting is more subtle than near-zero reconstruction (training) error, and near-zero reconstruction error also does not imply overfitting. This is consistent with the literature on benign overfitting (Bartlett et al., 2020).

Our assertion that reconstruction or training error cannot in general predict generalization error contradicts some existing claims in the literature. In particular, Jacot et al. (2006) derived a fixed relationship between training error and generalization error under restrictive assumptions of a Gaussian Process and in a large-n limit. Our results in Sect. "4.1.3" show empirically that generalization error can be much worse than is predicted by a fixed scaled version of training errors, *i.e.* $\frac{\theta^2}{z^2}\mathcal{E}_{re}$, especially when targets align with the trailing

Eigenvectors. To predict the generalization errors, Jacot et al. (2006) proposed the Kernel Alignment Risk Estimator (KARE)[2]:

$$
\begin{aligned}
\mathcal{E}_{ge} \approx \rho(z, n, y, K_n) &= \frac{\frac{1}{n} y^\top \left(\frac{1}{n} K_n + \lambda I_n\right)^{-2} y}{\left(\frac{1}{n} \mathrm{Tr}\left[\left(\frac{1}{n} K_n + \lambda I_n\right)^{-1}\right]\right)^2} \qquad ([2, pg.8]) \\
&= \lambda^2 y^\top (\frac{1}{n} K_n + \lambda I_n)^{-2} y \frac{1}{\frac{\lambda^2}{n} \mathrm{Tr}\left[\left(\frac{1}{n} K_n + \lambda I_n\right)^{-1}\right]} \\
&= \sum_{i=1}^{n} \left(\frac{y^\top u_i \lambda}{\frac{d_i}{n} + \lambda}\right)^2 \frac{1}{\frac{\lambda^2}{n} \mathrm{Tr}\left[\left(\frac{1}{n} K_n + \lambda I_n\right)^{-1}\right]} \\
&= \mathcal{E}_{re} \frac{1}{\frac{\lambda^2}{n} \mathrm{Tr}\left[\left(\frac{1}{n} K_n + \lambda I_n\right)^{-1}\right]} \qquad \text{Proposition 1} \\
&=: \mathcal{E}_{re} \cdot \phi(\lambda, X)
\end{aligned}
$$

where K is the Kernel/Gram matrix, λ is the ridge regularizer used by Jacot et al. (2006) (related to ours by $z = n\lambda$), ϕ is a function of z and X, y is the learning targets. Since $\mathcal{E}_{re}$ depends on the kernel K, and in turn, depends on input data X, $\mathbb{E}\mathcal{E}_{ge} \neq \mathbb{E}\mathcal{E}_{re} \cdot \mathbb{E}\phi(\lambda, X)$. This means expected risk cannot be written as a scaled version of expected training error; in fact, the gap between the true error and the error predicted by this scaling law is precisely the covariance between $\mathcal{E}_{re}$ and $\phi(\lambda, X)$. Moreover, Theorem 6 and Theorem 7 of Jacot et al. (2006) suggest that

$$
\begin{aligned}
|\mathbb{E}(\mathcal{E}_{ge}) - \frac{\theta^2}{z^2}\mathbb{E}(\mathcal{E}_{re})| &\leq |\mathbb{E}(\mathcal{E}_{ge}) - \tilde{R}^\epsilon| + \left|\frac{\theta^2}{z^2}\mathbb{E}(\mathcal{E}_{re}) - \frac{\theta^2}{z^2}\frac{z^2}{\theta^2}\tilde{R}^\epsilon\right| \\
&\leq \left(\frac{z^2 C + \theta^2}{z^2 n} + P_2\left(\frac{\mathrm{Tr}(T_K)}{z n^{1/2}}\right) + \frac{\theta^2}{z^2} P_3\left(\frac{\mathrm{Tr}(T_K)}{z n}\right)\right)\tilde{R}^\epsilon
\end{aligned}
$$

where θ is the signal capturing threshold, P_2, P_3 are polynomials with nonnegative coefficients, such that $P_2(0) = P_3(0) = 0$, T_K is the covariance operator, $\tilde{R}^\epsilon$ is also an approximator of expected risk (generalization error) $\mathbb{E}(\mathcal{E}_{ge})$. For finite n, the bound on the right may be non-negligible: there is no guarantee that that the polynomials P_2, P_3 are close to zero. This coincides with our analysis in Proposition 2, in contrast to the original claim from Jacot et al. (2006), that the training errors will deviate from generalization errors when generalization errors are far from zero. In other words, training errors alone cannot predict generalization errors except in regimes where generalization error is extremely low. Our experiments in Sect. "4.1.3" also validate this observation.

[2] We have adjusted the original notations for consistency and brevity, *e.g.* we use $\mathcal{E}_{ge}$ for $R^\epsilon\left(\hat{f}_z^\epsilon\right)$, and $\mathcal{E}_{re}$ for $\hat{R}^\epsilon\left(\hat{f}_z^\epsilon\right)$ in Jacot et al. (2006). We also omitted the arguments in these functions to avoid clutter. We refer the readers to the original paper for details.

3 Related Work

Cristianini et al. (2001) proposed the idea of kernel alignments, suggesting that when the kernel aligns with the labels, the model will have better generalization performance. They provide a measure of alignment based on the kernel and the learning target directly without providing Eigenvector-based analysis on generalization behavior. Jacot et al. (2006) shows that kernel alignments provide critical insights into the expected risk of the model. In particular, they proposed the concept of a signal capture threshold and demonstrated the relationship between training error and expected risk in closed form. Furthermore, while Jacot et al. (2006) did not make assumptions about the data, they made an implicit Gaussian assumption for the sampling operator. Simon et al. (2021) also proposes a similar Eigenlearning framework and introduces the concept of a conservative law of learning, suggesting that the learnability of the model is equal to the number of observations. Canatar et al. (2020) utilizes replica methods to study the spectral bias of kernel models and their application in neural networks. Canatar et al. (2023) extends the spectral theory on kernel models to study the connection between the encoding models of neural responses and the geometric properties of deep neural network (DNN) representations and neural activities. Also utilize replica methods, Bordelon et al. (2020) derives the generalization bounds for kernel regression as a function of the number of training samples. Barzilai and Shamir (2023) points out that much of the existing literature uses impractical assumptions that do not hold in practice. However, the bound provided by Barzilai and Shamir (2023) rely on choosing an arbitrary parameter k, and many of the results are only bounded by an order of magnitude.

The literature also examines the generalization of neural networks through kernel alignments. Canatar et al. (2020) analyzes the spectral bias of neural networks and demonstrates that the neural network training process can be decomposed using the Eigenfunctions of the neural tangent kernel. Canatar et al. (2020); Simon et al. (2021); Bordelon et al. (2020) also connects the kernel alignments to the neural network through the wide neural network limits and the neural tangent kernel. Wei et al. (2022) shows that the classical generalized cross-validation estimator can be extended to the overparameterized regime, such as in neural networks, through the framework of random matrix theory.

Recent studies, *e.g.* Hastie et al. (2022); Liang and Rakhlin (2018), show that high-dimensional ridgeless regression models can interpolate while still exhibiting good generalization. Hastie et al. (2022) also shows that the optimal ridge/regularization also vanishes in this regime, which is similar to our observations.

Our analysis focuses on the relationship between generalization and the estimation of Eigenvalues and Eigenvectors. Many efforts have been put into studying the Eigenvalue and Eigenvectors of the covariance matrix, *e.g.* Loukas (2017); Silverstein (1990); Mestre (2008), etc.

4 Experiments

In this section, we demonstrate our main findings through empirical experiments.

4.1 Eigen Alignments

4.1.1 Simulated Data

We first demonstrate that reconstruction errors in high dimensions can be arbitrarily small, as suggested in Proposition 2, with any bias attributable to the regularization parameter z. To this end, we randomly sample input data X with a dimension of $p \times 1000$. We fix the number of observations $n = 1000$, and vary the dimension of the input data X; specifically, $p = 10, 100, 200, 300, 500, 600, 800, 900$. For the learning objectives y, we utilize the normalized weighted sum, with random weights, of $1st$, $2nd$, $100th$, $400th$, $500th$ Eigenvectors of the linear kernel matrix $X^\top x$. We calculated the sum of the squared reconstruction error

$$n\mathcal{E}_{re} = \sum_i (\hat{y}_i - y_j)^2.$$

as our measurement. We plot the reconstruction errors in Fig. 2. It shows that as the dimension increases, we can achieve a near-zero reconstruction error, influenced by the bias introduced by the regularization parameter z. Notice the sharp decline that occurs after the dimension of the input data $p \geq 500$. In the case of our choice y, since it relies on the trailing Eigenvectors, it is unlikely to generalize well. However, a zero reconstruction error does not necessarily indicate poor generalization or overfitting, as we demonstrate in the following sections.

Next, we demonstrate the effects of regularization in non-parametric KRR; we randomly sample 1000 data points from a uniform distribution from $(-3, 3)$, *i.e.* each data point is scalar in $(-3, 3)$. Then, we use an RBF kernel to construct the kernel matrix and calculate the Eigenvalues and Eigenvectors. For training purposes, we randomly sampled 100 data points from the 1000 available data points. We use each Eigenvector we calculated (also sampled to 100 points) as the learning targets for training. To see how both learning targets and regularization affect generalization, we used trained models to predict the entire data set and calculate the generalization error. Figure 3 shows the results of this experiment. To better present the difference, we plot the generalization error on a logarithmic scale. It is clear from Fig. 3, that the generalization error increases as the rank increases, *i.e.* that the learning targets align with the trailing Eigenvectors. Additionally, we notice that for most of the learning targets, the larger the regularization, the larger the generalization error. This trend only reverses towards the trailing Eigenvectors. At the bottom of Fig. 3, we also show that large Eigen gaps also correspond to the smaller generalization errors.

Our experiments suggest that regularization is more effective when learning targets are aligned with trailing Eigenvectors. This coincides with our argument in the previous section: if the regularization is effective, the model is already in the unstable estimation regime. This is particularly true in higher dimensions, as illustrated in the following section, specifically, in Fig. 5. Note, we only consider the magnitude, not the variance of the error, where the regularization might be beneficial; however, the detailed analysis on variance is left to future work.

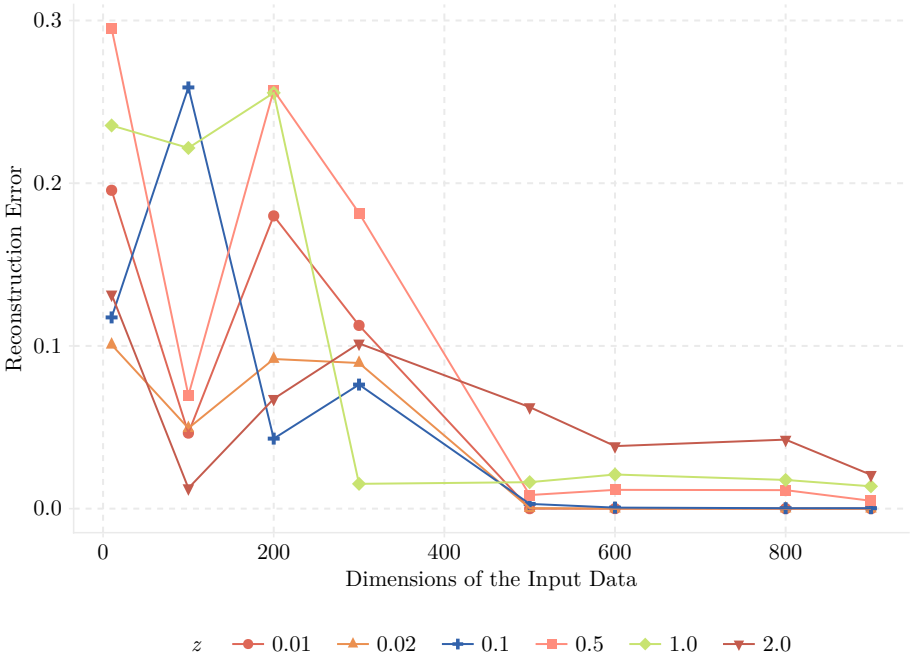


Fig. 2 Random data reconstruction error. We observe the following: (1) high reconstruction errors only appear when the dimension is not high enough (*i.e.* less than 500) for small regularizations; (2) when dimension $n = 600$, only the three highest regularization cases have reconstruction error significantly larger than zero

4.1.2 Real Datasets

Our experiments in this section utilize more realistic data to demonstrate our results, which are also applicable to real-world scenarios. For this experiment, we sampled 1000 images from the MNIST dataset, utilizing only those labeled 4 and 9. Despite the perceived simplicity, these data are actually of high dimensionality. We constructed a learning target based on the Eigenvectors of the kernel matrix using a squared inner product kernel $\mathrm{k}(x_i, x_j) = (x_i^\top x_j)^2$. For comparison, we construct two sets of learning targets: one aligned with the top Eigenvectors, $y_1 = u_3 - u_5$, where u_1 and u_3 are $3rd$ and $5th$ Eigenvectors, respectively; and one aligned with the trailing Eigenvectors, $y_2 = u_{100} + u_{101}$. The results are shown in Fig. 4.

On the left of Fig. 4, we observe that both training and testing error are quite low; this is shown by noticing the ground truth y_1 and predictions $\hat{y}_1$ are almost perfectly overlapped. In the right column of Fig. 4, we observe that despite the almost perfect overlap of the ground truth and the prediction for training, the performance on the tests is quite poor. These results show that alignment of the top Eigenvectors is the key to good generalization, and more importantly, in a nonparametric regime, the training error can be quite uninformative regarding the test performance.

To further evaluate this phenomenon, we compare the training errors and testing errors with an additional three datasets: CIFAR10, FashionMNIST, and SVHN2. To ease the com-

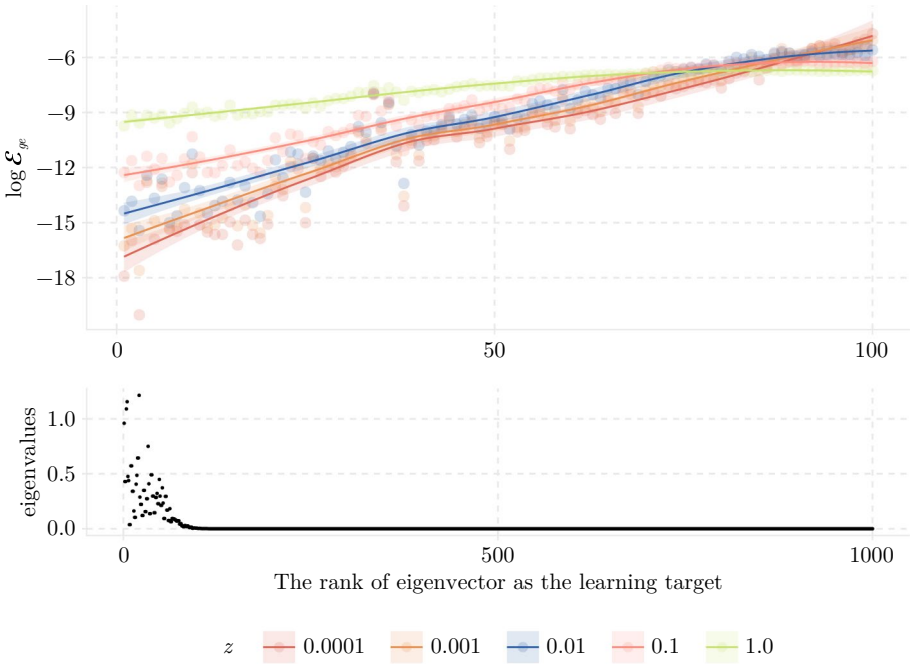


Fig. 3 Learning targets of the task affect the generalization performance. Learning targets aligned with top Eigenvectors (lower rank) tend to have lower generalization errors. Regularization tends to have a negative impact on generalization error for most learning targets, except those aligned with trailing ones (high-rank Eigenvector). Additionally, Eigen-gaps are largest

putational burden, we subset data by selecting 2 of the available labels, and we also limit the size of training samples to 1000 for each dataset. We evenly chose 10 Eigenvectors and multiplied them by 10 (to increase the magnitude) as the learning target for each run and computed the training and testing errors. The results are shown in Fig. 5.

In addition to the inner product kernel, we also utilized a Gaussian kernel to compare empirical behaviors of KRR. Specifically, we observe that the Eigenvalue distribution for a Gaussian kernel is almost uniform, indicated by the small Eigengaps $\min_j |d_i - d_j|$ for all Eigenvalues. This is expected since the L_2-norm-based kernel has an asymptotic rank of one when excluding the diagonal. As our analysis shows, the small Eigengaps cause significant prediction variance. Since the learning targets are small in norm, the model can accidentally achieve a small error in the testing set by simply making small predictions. Meanwhile, the inner product kernel shows much more stable prediction; additionally, as the Eigengaps decrease, the generalization errors increase.

In both cases, the training errors are relatively small across learning targets. Since the trailing Eigenvectors are smaller, the bias caused by the regularizations is also more significant in the trailing Eigenvectors. We see that despite the training error gradually increasing as the learning targets move to the tail of the Eigenspectrum, they remain close to zero. As analyzed in Proposition 2, the training error in non-parametric KRR can be arbitrarily small, up to regularization effects (mainly the relative magnitude of Eigenvalues compared to the

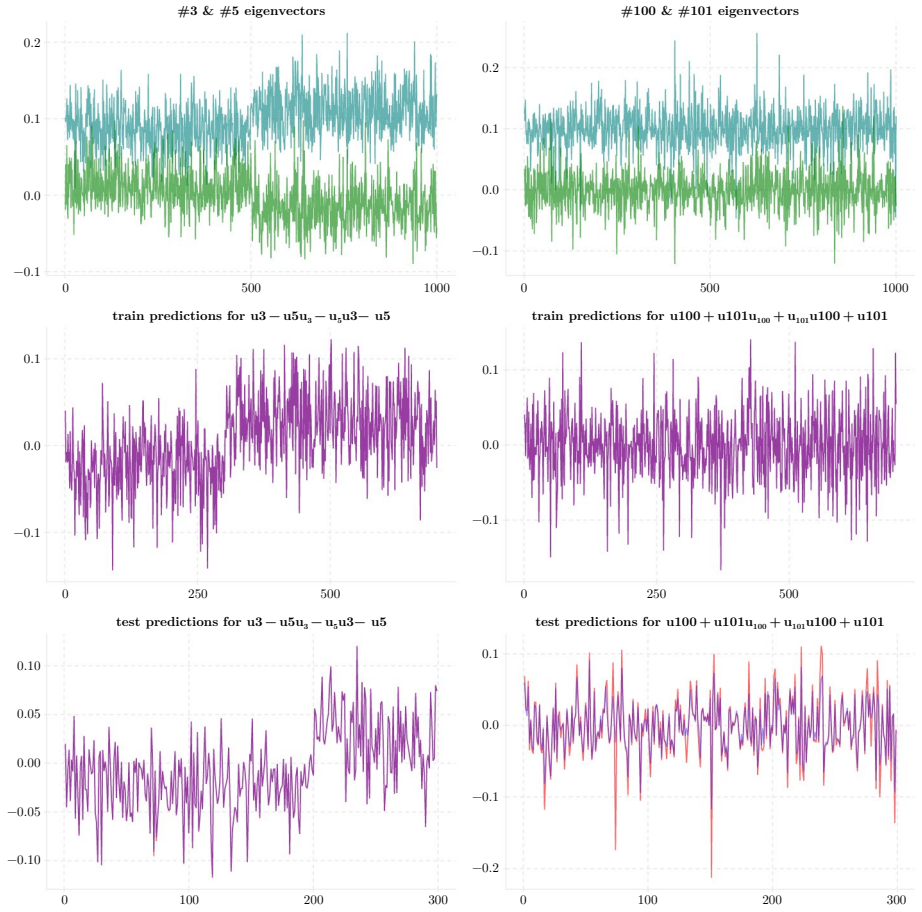


Fig. 4 With MNIST images data (labeled 4 and 9), we construct targets with top and trailing Eigenvectors. On the left, we see both training and testing have near-perfect performance (the blue prediction line overlaps with the red ground truth). On the right, we notice that although the training error is near zero, *i.e.* near-perfect overlapping of red and blue lines, the testing error is significant

regularizer) and numerical error. The testing errors, however, increase significantly for the trailing Eigenvectors.

Remark 9 We notice that in both Fig. 3 and left-hand sides of Fig. 5 the generalization error exhibit similar trends, despite the fact that the model in Fig. 3 also uses a Gaussian kernel. However, since it is in low dimension (dim = 1 in Fig. 3), the Gaussian kernel matrix has a more non-trivial Eigenvalue distribution: after subtracting the diagonal, the kernel matrix in Fig. 3 actually has a higher rank than the Gaussian kernel matrices in Fig. 5. Interestingly, the Eigenvalue distributions of a Gaussian kernel for low-rank data actually resemble the Eigenvalue distributions of inner product kernels, which have a quick decay in the magnitude of Eigenvalues. This is aligned well with our theory, which identifies large Eigengaps as a factor that can lead to better generalization.

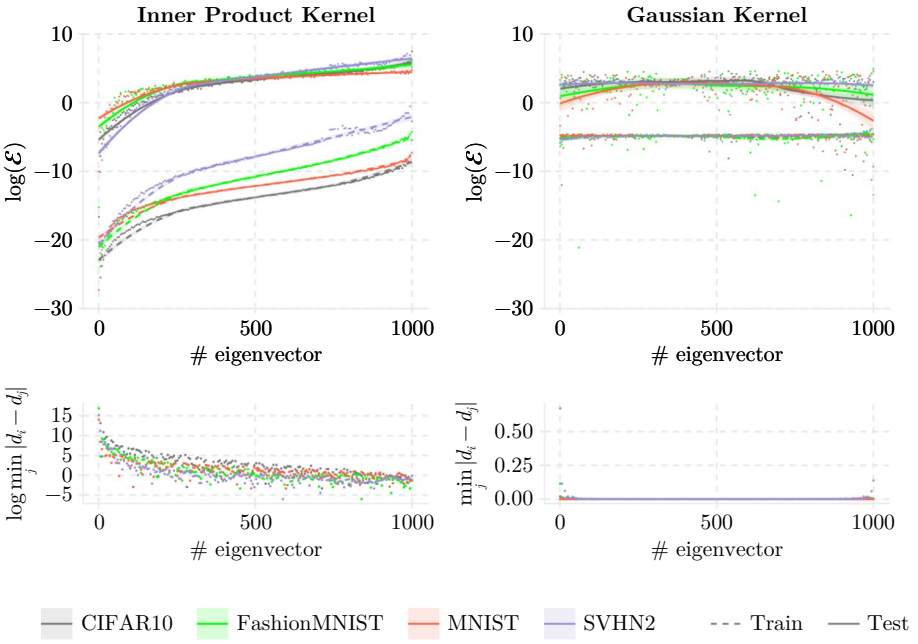


Fig. 5 Training error and testing error comparison for different kernel types and Eigenvectors as the learning targets. The Gaussian kernel showing a wider spread implies greater variance due to almost uniform Eigenvalue distributions. Both kernels exhibit a correlation between the generalization error and the Eigengaps. We used a fixed regularizer of $z = 0.2$ in all the experiments

Finally, we also observe the similar "flipping" effects on both Fig. 3 and the left-hand sides of Fig. 5: the regularization that causes the most bias in the top Eigenvector regime, leading to the most improvement in generalization in the trailing Eigenvector regime. This further demonstrates the limited effects of regularization in non-parametric KRR, such that regularization is most likely to be beneficial only when the learning targets are aligned with trailing Eigenvectors.

4.1.3 Empirical Bound

To further validate our generalization bound, we empirically demonstrate that our simple empirical estimator, as shown in eq.(12), is capable of capturing generalization behavior of KRR.

In the experiments, we compare our estimator with the Kernel Alignment Risk Estimator ρ from Jacot et al. (2006), and the test MSE estimator from Simon et al. (2021) in terms of predicting the sum-of-squared errors (SSE) for testing data (MNIST dataset and FashionMNIST dataset, same setup as Fig. 5), as these are most comparable to our work. We adjust all the formulas accordingly to calculate the sum of squared errors of the testing data points. The results are shown in Fig. 6. Additionally, we also plot $\frac{\theta^2}{z^2}\mathcal{E}_{re}$ (where θ is the signal capturing threshold proposed by Jacot et al. (2006)), the fixed relation between the reconstruction error (training error) and the test error. Unsurprisingly, as we have suggested,

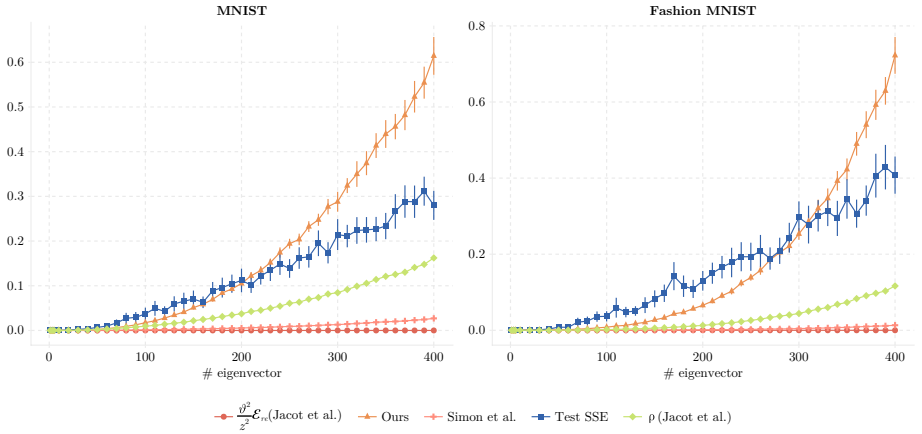


Fig. 6 True testing error compared with existing metrics from Jacot et al. (2006); Simon et al. (2021). Notice that the estimated error ρ from Jacot et al. (2006), and the scaled reconstruction error $\frac{\theta^2}{z^2}\mathcal{E}_{re}$ are diverging when the learning targets are aligned with trailing Eigenvectors. Our proposed simple estimator captures this Eigen-alignment behavior most closely, especially in the top Eigen-directions. Existing literature significantly underestimates prediction errors in trailing Eigen-directions. Eigen learning targets were computed using a subset of 1,000 images, and performance metrics were estimated by resampling the training set 100 times. The plot reports the mean, with error bars indicating ±1 standard deviation across random data splits

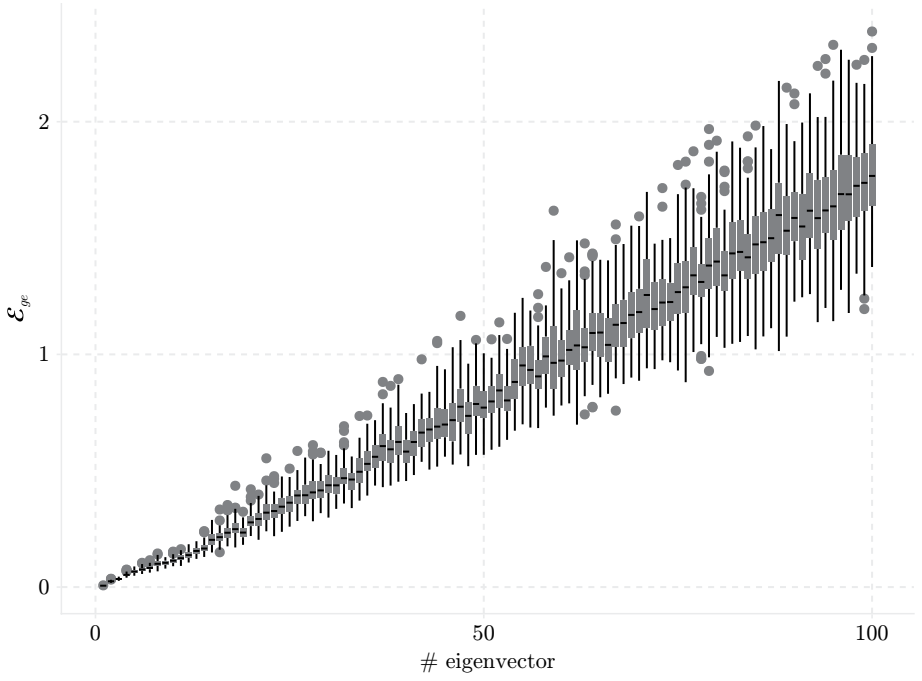


Fig. 7 Generalization error when treating each Eigenvector as an individual synthetic learning target

the training error has little predictive power over testing error. We observe that Simon et al. (2021) fails to track the risk in trailing Eigendirections at all. Jacot et al. (2006) also fails to track the testing errors for the FashionMNIST dataset. Our simple estimator is able to track the magnitude of the Test SSE most closely for the leading Eigenvectors (roughly up to index 250 for MNIST and index 300 for Fashion MNIST).

Remark 10 It is worth emphasizing that our goal is not to propose an accurate estimator, since we prioritize brevity over tightness in the analysis. Nonetheless, we find that the simple estimator above can still track the test error with reasonable accuracy. Therefore, our results show a significant limitation in current literature regarding generalization errors in trailing Eigen-directions. Note that the estimator begins to overestimate the errors in the trailing Eigendirections; this is expected because Eigengap-based bounds are known to be loose for the corresponding trailing Eigenvectors. Our theoretical results remain valid, and sharper Eigenvector perturbation bounds could be used to construct an improved estimator, which we defer to future work.

We also show in Fig. 7 that our analysis indicates that when learning targets are aligned with the trailing Eigenvector, not only does the generalization error increase, but the volatility also increases. In the experiment, we randomly sampled 700 images from a total of 100 to form our training set and utilized the remaining images as the testing set. We repeat each learning target 100 times to illustrate its volatility.

4.2 Reconstruction Error and Generalization

Next, we will demonstrate that, according to Corollary 1, the reconstruction error has limited predictive power regarding the generalization error. We sampled 1000 images labeled "4" or "9" from the MNIST dataset for our training data. By choosing the following four kernels, we demonstrate the following: (1) an improperly scaled kernel can have a small reconstruction error due to the magnified Eigenvalues; (2) A near-zero training error does not indicate overfitting, as a squared kernel can exhibit both a small reconstruction error and a generalization error; 3) trivial kernels can exhibit near-zero training errors while demonstrating trivial generalization. Without loss of generality, all the kernels used are inner-product kernels. The first three kernel functions are as follows:

$$f_1(t) = t, \quad f_2(t) = \frac{t}{\sqrt{784}}, \quad f_3(t) = \frac{t^2}{\sqrt{784}}. \tag{13}$$

Then, the (j, k)th element of the kernel matrix constructed with f_i is $[K_i]_{j,k} := f_i(x_j^\top x_k)$. The trivial kernel[3] is defined as $[K_4]_{j,k} = 500\delta_{jk}$, where δ is the Dirac delta. We use $1/n \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$ to calculate both the reconstruction error and the generalization error. All kernels were trained and tested using the same data set.

[3] Although this is not a valid kernel in RKHS, but in finite dimension, it is a valid way to construct kernel matrices.

Table 2 Reconstruction error and generalization error comparison for different kernels

	Kernels			
	K_1	K_2	K_3	K_4
Reconstruction error	0.08	0.106	4.6e−6	4.0e−8
Generalization error	0.382	0.279	0.193	1.99

From Table 2, we observe the following: (1) the improperly scaled kernel K_1 exhibits a smaller reconstruction error in comparison to the properly scaled counterpart K_2, due to larger Eigenvalues; however, it generalizes worse; (2) K_3 exhibits both a smaller reconstruction error and a generalization error, suggesting that a near-zero reconstruction error does not indicate overfitting or poor performance; (3) K_4 also exhibits a near-zero reconstruction error due to high Eigenvalues; however, it demonstrates poor generalization. This indicates that reconstruction is a trivial task with minimal predictive power in generalization.

5 Discussion: Limitations and Comparison

One limitation is that we assume that observed labels y are identical to the ground truth; that is, we assume no noise in the targets. Based on our analysis, one may be able to bound the effects of (either homoskedastic or heteroskedastic) noise by considering the projection of the noise onto (mis)estimated Eigenvectors. A second limitation is that our analysis studies the problem of learnability in terms of observed information, though in real-world settings data may contain missing values. It is possible that the standard practice of data imputation (using a kernel) could be incorporated into our analysis, *e.g.* as additional structured noise in the perturbed ΔK_n. A third limitation is that our generalization bound in eq.(11) was derived from a sequence of inequalities, which may not be tight in practice; in this sense, our results are an initial exploration of how perturbation analysis can be applied to KRR, and we expect that further developments in perturbation analysis will be able to sharpen these bounds. All of these limitations point to interesting directions for future work.

Our work differs from several notable existing studies, such as Jacot et al. (2006); Canatar et al. (2020); Bordelon et al. (2020); Simon et al. (2021), in terms of the assumptions and perspectives that underpin our analysis. We primarily focus on the effects of finite sample sizes on the generalization error, while the existing literature addresses asymptotic behaviors in infinite dimensions, typically through Gaussian process-related tools. We aim to provide a fresh approach to kernel alignment analysis with finite matrix perturbation theory. As we have stated earlier, we prefer the brevity of the results to illustrate the qualitative insights our analysis can provide, where the tightness of our bounds is likely to be further refined. However, we will defer the sharpening of the bound, along with further investigation into more structured kernel matrices, to future work.

Existing work on the analysis of generalization, *e.g.* Smale and Zhou (2005); Simon et al. (2021); Jacot et al. (2006), tends to treat the function as an infinite-dimensionall vector, with the training data viewed as random samples drawn from a Gaussian process. However, this assumption overlooks the complexity and richness of the information embedded within the training data and the learning targets. We found that generalization fundamentally relies on the consistent estimation of "useful" information, *i.e.* Eigenvalues and Eigenvectors that are aligned to the targets, particularly when dealing with finite sample sizes. In contrast, in Jacot et al. (2006), the generalization error and the training error are correlated by a factor $\theta(\lambda^2)/\lambda$

, which is independent of the learning targets. We have demonstrated, through both our analysis and numerical experiments, that this is not the case; we utilize synthesized learning targets to demonstrate that the ratio of generalization error to training error depends strongly on the learning targets. Simon et al. (2021) also provided a similar ratio called "overfitting coefficient", which suffers from a similar issue.

Our results also provide insights into the limitations of regularization for Kernel Ridge Regression (KRR) from the perspective of a finite kernel matrix. From our previous analysis and Remark 2, we show that $d_i + z \ll \mathbb{P}^y_{u_i} d_i$ is necessary for the Eigenvector to be regarded as relevant. Additionally, the Eigenvalues of the top Eigenvectors typically increase as the size of the kernel matrix grows, *i.e.* with an increase in the number of observations. This implies that a higher regularization is more permissible. The small differences induced by regularization tend to increase as the learning targets align with the trailing Eigenvectors. These observations are aligned with Bordelon et al. (2020), who suggest that explicit regularization might hinder performance when regularization is large relative to the sample size.

Finally, like Simon et al. (2021), our work differs from that of Canatar et al. (2020) and Bordelon et al. (2020) in the sense that we derived an explicit expression for the generalization error, *i.e.* without relying on implicit expressions. In our case, this comes from using a matrix perturbation perspective rather than random matrix theory as used by Jacot et al. (2006), allowing us to make statements about the finite sample regime. As we have shown in the Sect. "4.1.3", our results reveal that existing literature significantly underestimates the generalization errors in trailing Eigen-directions.

6 Conclusion

Our study departed from the conventional functional space view, focusing instead on how the specific choice of learning targets significantly impacts a model's generalization behavior, particularly in the context of Kernel Ridge Regression (KRR). This approach allows us to meticulously dissect the intricate relationships among reconstruction error, overfitting, and robust generalization. Our findings show that generalization depends on the model's ability to extract information from data consistently, and different learning targets can lead to significant differences in generalization performance. In contrast to previous work, we observed that the reconstruction error often has minimal predictive power for generalization, as a near-zero training loss is nearly universally achievable. We demonstrate this using synthesized learning targets: as we change the learning targets, the generalization error changes drastically, while the reconstruction error remains relatively constant. This is because the reconstruction error is independent of the stability of the information. In Kernel Ridge Regression (KRR), this phenomenon arises from the perfect alignment of the Eigenvectors of two kernel matrices in the KRR prediction formula $\hat{y} = K'KA$.

In passing, we also notice that the regularization parameter frequently has a very limited positive impact in non-parametric KRR, as we show in the analysis and experiments. Regularization is more likely to have a positive impact when the learning targets align with trailing Eigenvectors, but aligning with trailing Eigenvectors is already a sign of poor generalization. We proposed bounding the generalization error through the lens of Eigenvalue and Eigenvector estimation. We show that good generalization requires either increasing alignment of Eigenvectors or an increase in the Eigenvalue; both of these can be achieved

if the learning targets are aligned with the top Eigenvectors. Based on our generalization error bound, we are able to derive a simple empirical risk estimator, which we have shown to track generalization errors better than existing literature in trailing Eigen-directions.

Author Contributions Y.L. conducted the primary research work; E.F. have participated weekly discussion on the topic; R.L. provided some valuable insights and suggestions, and helps with revision of the manuscript; D.K. oversaw the overall project. All authors reviewed the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Bartlett, P. L., Long, P. M., Lugosi, G., & Tsigler, A. (2020). Benign overfitting in linear regression. *Proceedings of the National Academy of Sciences of the United States of America, 117*(48), 30063–30070.

Barzilai, D., & Shamir, O. (2023). Generalization in kernel regression under realistic assumptions.

Bordelon, B., Canatar, A., & Pehlevan, C. (2020). Spectrum dependent learning curves in kernel regression and wide neural networks. *International Conference on Machine Learning, 119*, 1024–1034.

Canatar, A., Feather, J., Wakhloo, A., & Chung, S. (2023). A Spectral Theory of Neural Prediction and Alignment. In: Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., Levine, S. (eds.) *Advances in Neural Information Processing Systems*, vol. 36, pp. 47052–47080. Curran Associates, Inc.

Canatar, A., Bordelon, B., & Pehlevan, C. (2020). Spectral bias and task-model alignment explain generalization in kernel regression and infinitely wide neural networks. *Nature Communications, 12*, 2914.

Couillet, R., & Liao, Z. (2021). Random matrix methods for machine learning: When theory meets applications

Couillet, R., & Benaych-Georges, F. (2016). Kernel spectral clustering of large dimensional data. *Electronic Journal of Statistics, 10*(1), 1393–1454.

Cristianini, N., Shawe-Taylor, J., Elisseeff, A., & Kandola, J. (2001). On kernel-target alignment. *Advances in Neural Information Processing Systems,14*

Cucker, F., & Zhou, D. X. (2007). Learning theory: An approximation theory viewpoint. *Cambridge monographs on applied and computational mathematics.* Cambridge: Cambridge University Press.

Davis, C., & Kahan, W. M. (1970). The rotation of eigenvectors by a Perturbation. III. *SIAM Journal on Numerical Analysis, 7*(1), 1–46.

Eldridge, J., Belkin, M., & Wang, Y. (2017). Unperturbed: spectral analysis beyond Davis–Kahan. *International Conference on Algorithmic Learning Theory* (pp. 321–358). arxiv.abs/1706.06516

Fan, J., Wang, W., & Zhong, Y. (2018). An l∞ Eigenvector perturbation bound and its application to robust covariance estimation. *Journal of Machine Learning Research: JMLR, 18*, 207.

Hastie, T., Montanari, A., Rosset, S., & Tibshirani, R. J. (2022). Surprises in high-dimensional Ridgeless least squares interpolation. *Annals of Statistics, 50*(2), 949.

Jacot, A., Simcsek, B., Spadaro, F., Hongler, C., & Gabriel, F. (2020). Kernel alignment risk estimator: Risk prediction from training data. *Advances in Neural Information Processing Systems, 33*, 5568–15578.

Lancewicki, T. (2019). Kernel matrix regularization via shrinkage estimation. *Advances in Intelligent Systems and Computing* (pp. 1292–1305). Springer, Cham.

Liang, T., & Rakhlin, A. (2018). Just interpolate: Kernel "Ridgeless" Regression can generalize. arXiv [math. ST] (2018) [math.ST]

Liao, Z., & Couillet, R. (2019). Inner-product kernels are asymptotically equivalent to binary discrete kernels.

Loukas, A. (2017). How close are the eigenvectors of the sample and actual covariance matrices? *International Conference on Machine Learning* (pp. 2228–2237)

Martin, C.H., & Mahoney, M.W. (2020). Heavy-tailed universality predicts trends in test accuracies for very large pre-trained deep neural networks. In: *Proceedings of the 2020 SIAM International Conference on Data Mining*, pp. 505–513. Society for Industrial and Applied Mathematics, Philadelphia

Mestre, X. (2008). Improved estimation of eigenvalues and eigenvectors of covariance matrices using their sample estimates. *IEEE Transactions on Information Theory, 54*, 5113–5129.

Mestre, X. (2008). On the asymptotic behavior of the sample estimates of eigenvalues and eigenvectors of covariance matrices. *IEEE Transactions on Signal Processing: A Publication of the IEEE Signal Processing Society, 56*(11), 5353–5368.

Muandet, K., Sriperumbudur, B., Fukumizu, K., Gretton, A., & Schölkopf, B. (2016). Kernel mean shrinkage estimators. *Journal of Machine Learning Research: JMLR, 17*(48), 1–41.

Silverstein, J. W. (1990). Weak convergence of random functions defined by the eigenvectors of sample covariance matrices. *Annals of Probability, 18*(3), 1174–1194.

Simon, J.B., Dickens, M., Karkada, D. & DeWeese, M.R. (2021). The Eigenlearning framework: A conservation law perspective on kernel regression and wide neural networks. arXiv [cs.LG]

Smale, S., & Zhou, D.-X. (2005). Shannon sampling II: Connections to learning theory. *Applied and Computational Harmonic Analysis, 19*(3), 285–302.

Wei, A., Hu, W., & Steinhardt, J. (2022). More than a toy: Random matrix models predict how real-world neural representations generalize. *International Conference on Machine Learning, 06176*, 23549–23588.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Yang Liu[1] · Ernest Fokoue[2] · Richard Lange[1] · Daniel Krutz[1]

✉ Yang Liu
yl4070@rit.edu

Ernest Fokoue
epfeqa@rit.edu

Richard Lange
rdlvcs@rit.edu

Daniel Krutz
dxkvse@rit.edu

[1] Golisano College of Computing and Information Sciences, Rochester Institute of Technology, Rochester, NY, USA

[2] College of Science, Rochester Institute of Technology, Rochester, NY, USA